

This talk is a practitioner's tour of the Catapult HLS user-view scheduling rules and the formal architecture behind them.

The audience is hardware architects and design verification engineers. You do not need to be an HLS expert or a formal-methods specialist. The goal is that you leave knowing: what the contract lets HLS do to your design, what it forbids, how to write source models and testbenches that stay inside it, and what the formal guarantees actually say - and do not say.

Theorem and appendix names appear in small print so you can find the precise statement in the rules document, but the talk keeps the formal vocabulary to a minimum.

EXECUTIVE SUMMARY

A small contract that lets verification stay at the source

HLS may add latency, pipeline loops and reorder memory accesses. The scheduling contract states exactly what it may not change — so a testbench written for the SystemC model stays meaningful for the RTL.



The goal

Predictable source-to-RTL behavior: verify and debug mostly at the source, with one testbench for both models.



The rules

Three classes of IO — message passing, signal IO and synchronization — each with one uniform rule.



The freedom

Pipelining, added states, late sampling of stable inputs, permitted memory reordering, finite buffering.



The guarantees

With the required evidence: proved source interface properties hold on RTL, and the required workload completes.



Your part

Latency-insensitive interfaces, well-anchored signal IO, explicit memory, reset and channel structure.

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Five ideas carry the whole talk.

1. The goal is predictable source-to-RTL behavior: most verification and debug happen on the pre-HLS SystemC model, and one testbench serves both the model and the RTL.
2. The rules are simple. There are three classes of IO operation - message passing, signal IO and synchronization - and each is scheduled by one uniform rule.
3. Simplicity does not cost QOR: HLS keeps the freedom to pipeline, add states, sample stable inputs late, reorder memory accesses that are proven conflict-free, and implement finite buffering.
4. Two formal results back the rules: a safety result (covered RTL behavior is explained by the reference model) and a completion result (required outputs eventually arrive).
5. The architect's part is to choose interfaces and structures that make that evidence possible.



Four acts.

The why: what problem the contract solves, and the handful of terms you need.

The rules: the basic conceptual model, then the optimizations HLS may still perform inside it.

Modeling practice: how to write source models and testbenches that stay inside the contract.

The guarantees: what the formal results say, how they are built from local evidence, what they do not claim, and who supplies which evidence.

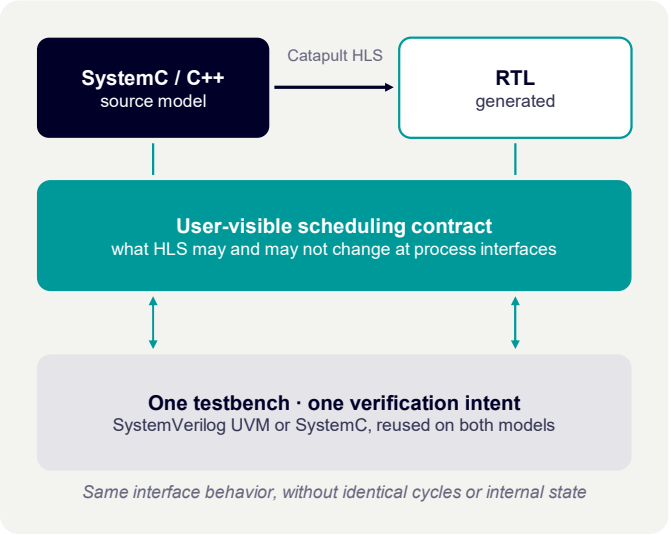
HLS pays off only if verification stays at the source

The problem

- ✕ RTL verification is among the most costly and time-consuming parts of a hardware flow.
- ✕ Generated RTL can differ substantially from the source model in structure and in timing.
- ✕ If teams must return to RTL as the primary level for verification and debug, HLS loses much of its advantage.

The aim

Designers and DV engineers reason primarily at the source level, while keeping a precise way to qualify the resulting implementation.



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Start with the economics. RTL verification is one of the most expensive and slowest parts of a hardware flow, and machine-generated RTL can look very different from the source in structure and in timing. If teams must return to that RTL as the primary level for functional verification and debug, much of the advantage of HLS is lost.

The scheduling contract is the bridge. It is user-visible and tool-neutral: it states what HLS may and may not change at process interfaces. That lets the same verification intent - for example a UVM or SystemC testbench - be reused across the source reference model and the RTL, without requiring the RTL to preserve every source cycle or internal state.

The testbench author should be able to rely on the interface contract without knowing the HLS tool’s internal scheduling.

An analogy from RTL synthesis

In Verilog

```
Out1 <= #some_delay new_val;
```

- Simulation honors the delay. Longer than a clock period, it changes cycle-level behavior.
- RTL synthesis ignores every delay — and may warn: “simulation and synthesis results are likely to mismatch.”
- Nobody expects synthesis to match the simulation exactly. Everyone aligns on the conceptual model.

	RTL synthesis	HLS with this contract
Conceptual model	Clocked RTL semantics	The user-view scheduling rules
Simulation model	Verilog simulation, including #delays	SystemC / C++ simulation of the source
Synthesis	Ignores delays; flags suspicious ones	Schedules within the rules; flags RULE 1 / RULE 2 violations

Why it works: the conceptual model is simple enough for users and tool vendors to align around, so small gaps between simulation and synthesis are easy to manage.

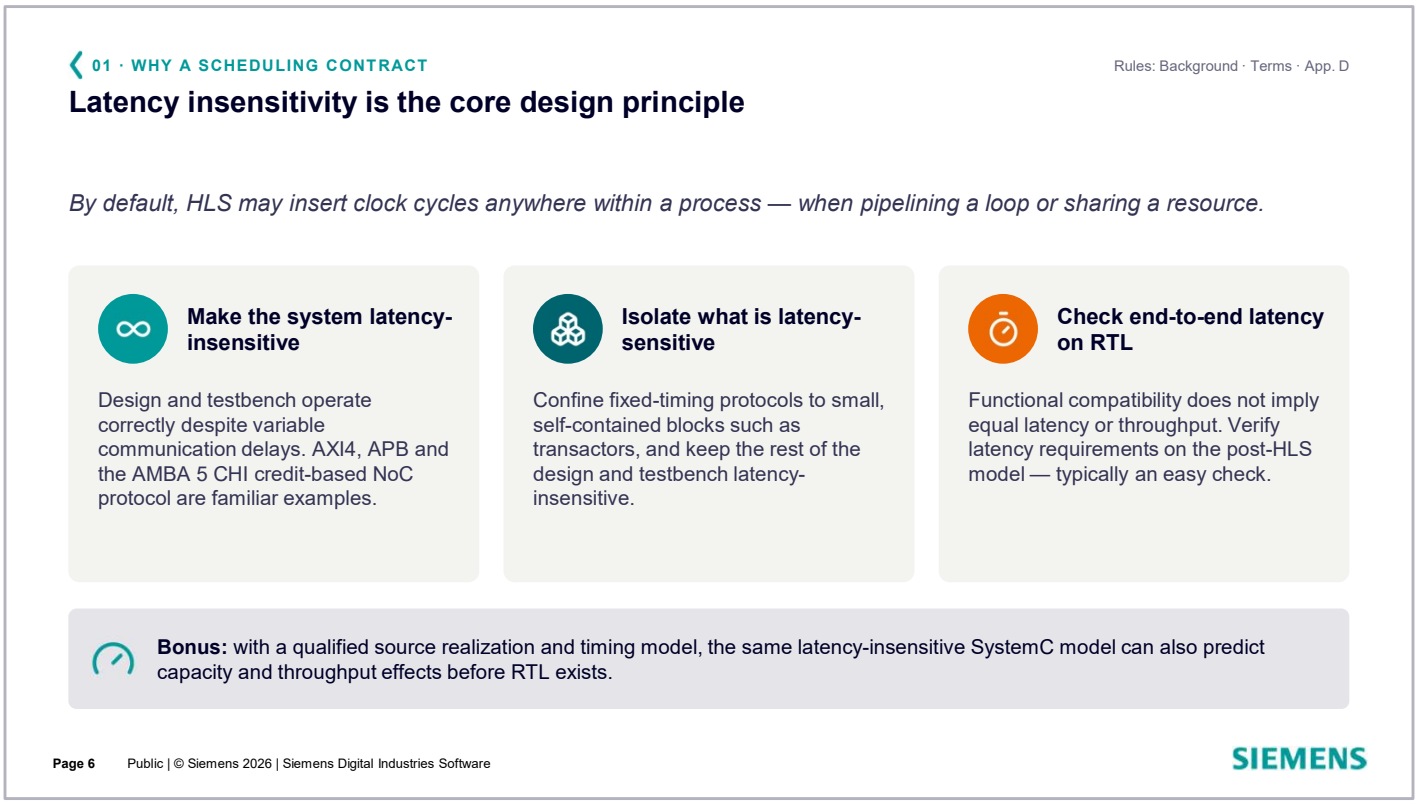
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The rules document opens with a familiar analogy. In Verilog, 'Out1 <= #some_delay new_val' simulates with the delay. If the delay is shorter than the clock period it probably does not change cycle behavior; if it is longer, it probably does. RTL synthesis ignores all delays anyway, and good tools warn that simulation and synthesis are likely to mismatch, or reject the model.

Nobody argues that RTL synthesis should match Verilog simulation exactly. The flow works because RTL is a simple conceptual model that users and tool vendors align around, and the small gaps between the simulation model and the conceptual model are easy to manage.

The scheduling rules play the same role for HLS. And the RULE 1 / RULE 2 diagnostics we will see later are the HLS counterpart of that 'delay greater than clock period' warning.



By default HLS may add clock cycles anywhere inside a process - to pipeline a loop or to share a multiplier. So the overall approach is to make the whole design and testbench latency-insensitive to the maximum extent possible, while still enabling the key hardware optimizations.

Latency-insensitive means the system works correctly despite variable communication delays. AXI4, APB and AMBA 5 CHI are familiar examples.

Where a protocol really is latency-sensitive, isolate it in a small self-contained block and keep the rest latency-insensitive.

End-to-end latency requirements still matter. The recommendation is to keep the latency-insensitive modeling style and check latency on the post-HLS model, which is usually easy. Functional trace compatibility alone does not establish latency or throughput equality - those need their own qualified timing model.

A factory analogy for the rules

People at workbenches	»	**Processes**
Their written instructions	»	**C++ code in a process**
Objects handed between people	»	**Messages (blocking or non-blocking)**
A wall clock everyone can see	»	**The hardware clock**
Flags, seen at the next tick	»	**Signals (delayed update)**
Subordinates doing subtasks	»	**Pipeline stages**
Meeting at a barrier	»	**Explicit synchronization**
Raising a reset flag	»	**Hardware reset**

Now assume

- Task and transfer times vary
- Everyone works as quickly as they can
- Several products — chairs in different colors and styles — are built at once

To produce furniture reliably, everyone must follow rules like those in this contract.

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Appendix A grounds the rules in a factory that builds wooden furniture.

People at workbenches are processes; their written instructions are the C++ code. They pass objects - messages - to each other, blocking or non-blocking. A wall clock that everyone can see is the hardware clock. Colored flags are signals, and a flag change is seen by others only at the next tick of the second hand: that is the delayed-update semantics of signals between sequential processes. Hiring subordinates to do subtasks is pipelining. Agreeing to meet at a barrier is explicit synchronization, and a reset flag restarts work.

Now add realistic assumptions: task and transfer times vary, everyone works as fast as they can, and the factory makes several products at once. To produce furniture reliably, everyone has to follow rules like the ones in this contract. The rules are not arbitrary; they are what predictable concurrent work needs.

Rules: Appendix A

In this example, overlap works; reversal can deadlock

```
// Person 1
seats.Push(s);
backs.Push(b);
```

```
// Person 2
s = seats.Pop();
b = backs.Pop();
```

Blocking calls, zero storage between people: a Push completes only when the matching Pop does.

Each transfer takes at least one minute.

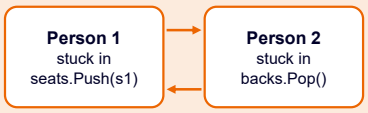
As written: 5 minutes

#	Person 1	Person 2
1	seats.Push(s1)	
2	backs.Push(b1)	s1 = seats.Pop()
3	seats.Push(s2)	b1 = backs.Pop()
4	backs.Push(b2)	s2 = seats.Pop()
5		b2 = backs.Pop()

Both overlap IO: 3 minutes

#	Person 1	Person 2
1	seats.Push(s1) backs.Push(b1)	
2	seats.Push(s2) backs.Push(b2)	s1 = seats.Pop() b1 = backs.Pop()
3		s2 = seats.Pop() b2 = backs.Pop()

Person 2 reverses: deadlock



Person 1 waits for a seats.Pop that never comes; Person 2 waits for a backs.Push that never starts.

Default rule: message calls may overlap, but their source issue order must be preserved. Completion needs separate evidence.

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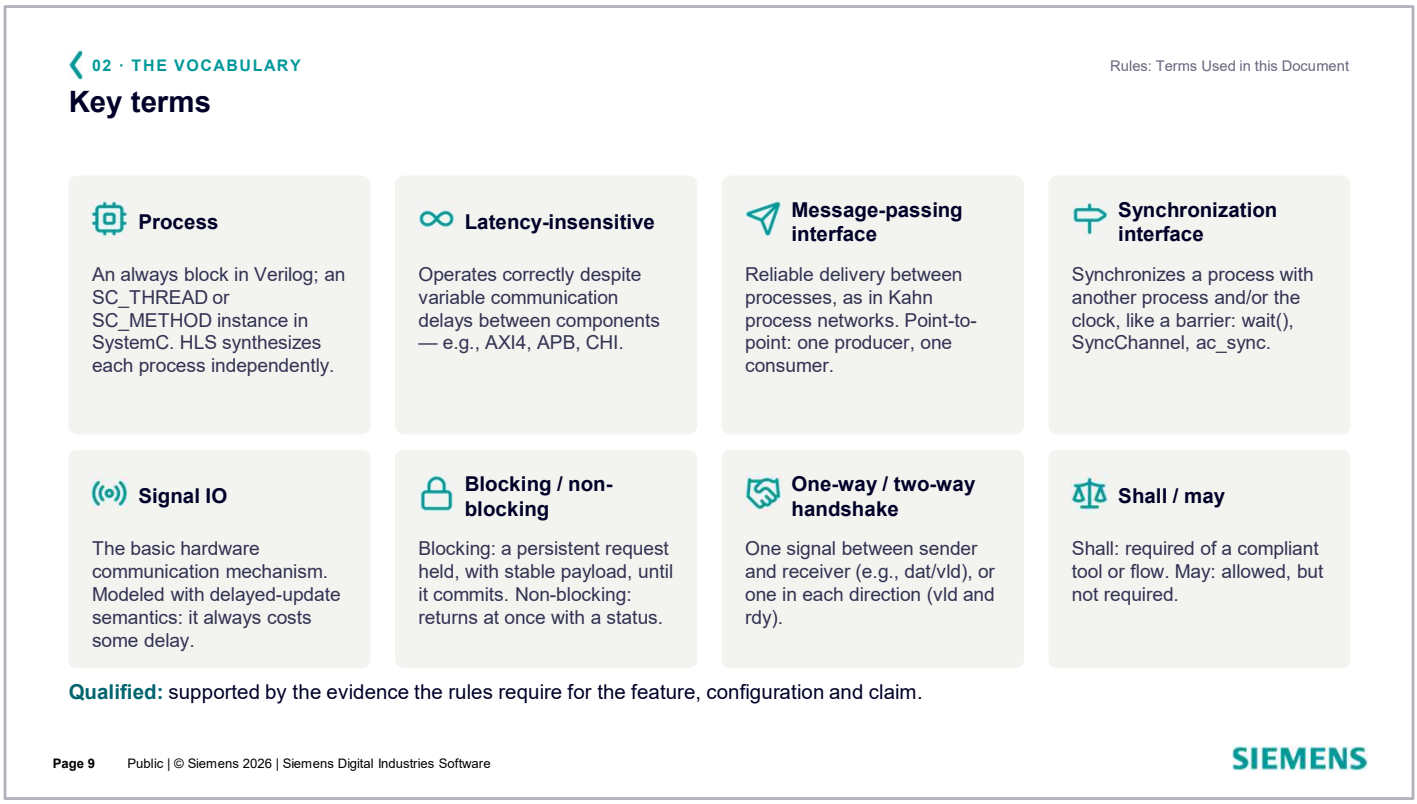
Walk through the tables in order. With the calls as written, Person 1 offers the first seat, then its back, then the second seat and back. The two workers' handoffs occupy the five numbered minutes shown. Each transfer takes at least one minute, and there is no storage between the workers.

When both workers overlap their independent seat and back calls while preserving issue order, the same work finishes in the three minutes shown. A later request may begin before an earlier one finishes.

Now reverse Person 2's calls. Person 1 waits to hand over a seat while Person 2 waits for a back. Neither reaches the matching call, so the first handoff cannot complete.

This example establishes a legal overlap and a specific deadlock caused by reversing issue order. It does not establish that every overlap is harmless. Finite storage, early requests, controllers and the environment still need their own safety and completion evidence.

(References: rules Appendix A, E3 and Appendix K.)



A few terms recur throughout.

A process is an always block in Verilog or an SC_THREAD / SC_METHOD instance in SystemC. HLS finds the processes and synthesizes each one independently.

Message-passing interfaces are the Kahn-process-network style of communication: reliable, ordered delivery. In this document every channel is point-to-point, with exactly one producer and one consumer. Shared resources are modeled with explicit arbiter processes.

Synchronization interfaces are like barriers - wait(), SyncChannel, ac_sync.

Signals always cost some delay; HDLs and SystemC model this with delayed-update semantics.

Blocking means a persistent request held until the operation commits; non-blocking returns immediately with a status. 'Shall' is required of a compliant tool or flow, 'may' is permitted.

Qualified means supported by the evidence the rules require for the feature, configuration and claim. It describes a specific scope of evidence, not a general certification of a tool or every design it can produce.

Three classes of IO operation

Message passing

ac_channel methods; Push, Pop, PushNB, PopNB (e.g., Connections In<> / Out<>)

Moves data between processes. May shift and overlap in time.

Synchronization

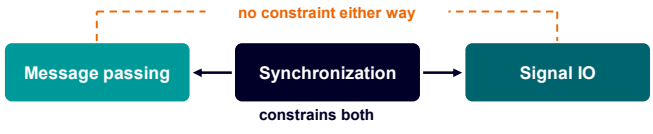
wait(), ac_wait, SyncChannel (sync_in / sync_out), ac_sync

Defines the boundaries the other operations respect. Always stays in source order.

Signal IO

sc_signal / sc_in / sc_out reads and writes; C++ direct inputs

Sequential IO: anchored to synchronization. Combinational IO: observed after settling.



Message calls never move a signal's anchor, and signal statements add no message-order constraints. No "protocol regions" are needed.

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Every scheduling rule talks about just three classes of IO operation.

Message passing: all ac_channel methods and the SystemC Push, Pop, PushNB and PopNB calls - for example Matchlib Connections In and Out ports.

Synchronization: SystemC wait and ac_wait, plus SyncChannel and ac_sync.

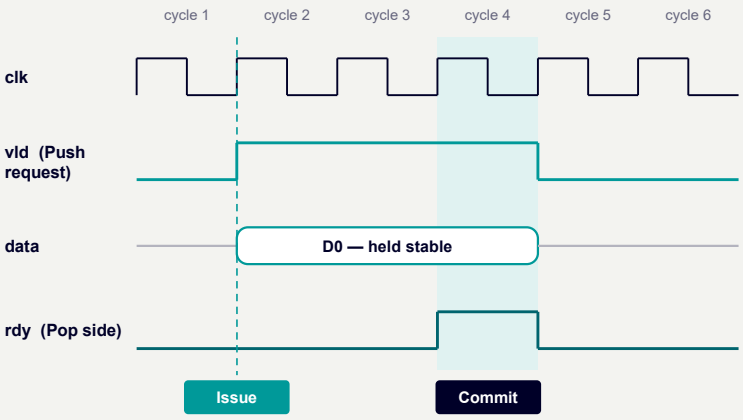
Signal IO: SystemC signal and port reads and writes, plus C++ model direct inputs.

The relationship between them is the key to the whole contract. Synchronization calls constrain both other classes: they anchor signal IO and fence message passing. But message passing and signal IO do not constrain each other's scheduling - a Pop placed between a wait and a signal read does not move that read. Signal values can of course still determine payloads and control flow.

These are the same operations verification engineers already use in testbenches, which is why the rules are easy to adopt. Note what is absent: no 'protocol regions' or similar tool-specific constructs.

Sequential-process signal accesses use synchronization anchors. For combinational processes, observe the stable values after the network settles, under the applicable combinational rules (R2C). Ordinary latency-insensitive interfaces do not themselves establish the selected proof profile.

Issue and commit: the two moments of a transfer



Blocking Push: valid and payload are held from issue until both valid and ready are high at the commit edge.

Issue

The cycle the process starts requesting: valid for a Push, ready for a Pop.

Commit

After signals settle, if valid and ready are both high at the applicable clock edge, the transfer must occur with the data sampled there.

Persistent requests

A blocking request stays asserted, payload stable, until commit or reset — no try-and-withdraw.

Registered results

A result returned at a cycle boundary cannot drive a dependent request in that same cycle.

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Issue is when the process starts requesting a transfer at its channel boundary: valid for a Push, ready for a Pop. Commit is when that transfer occurs. At an applicable clock edge, after signals settle, valid and ready both high requires a transfer using the data sampled at that edge (Core.TransferLaw).

Any grant, guard or resource restriction must already be reflected in the settled handshake signals and supporting state. It cannot cancel a transfer after both signals have been sampled high.

A blocking request persists until commit or an affecting reset. A Push also holds its payload stable. There is no try-and-withdraw behavior.

For sequential processes, a returned result cannot generate a dependent request in the same cycle. That request must issue in a later cycle (Core.RegSeq).

Requests on different interfaces may overlap while preserving source issue order. That does not impose a strict completion order between those interfaces. Also distinguish a physical movement through internal staging from a source-level issue or completion: the declared interface mapping must say which physical activity represents each logical event.

(References: Issue vs. Commit, Appendix C, Core.IC, Core.TransferLaw and Core.RegSeq.)

The basic conceptual model in four rules

1

Synchronization calls stay in source order.

wait(), SyncChannel and ac_sync are never reordered within a process.

2

Signal reads occur at the closest preceding synchronization call.

A read observes the value present at that anchor.

3

Signal writes occur at the closest succeeding synchronization call.

That is when the value becomes visible to others.

4

Message passing may shift, share a cycle or overlap…

…within the ordering and boundary constraints on the next slide.

Scope

Sequential processes without loop pipelining. Rolled loops may contain waits. Combinational IO uses settled observations.

Key idea

Everything HLS adds — extra states, pipeline stages, late sampling — must remain explainable by these four rules at the process interfaces.

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Here is the basic conceptual model. It covers processes with no loop pipelining; preserved loops are fine, and they may contain wait statements.

Rule 1: synchronization calls within a process always stay in source order.

Rule 2: signal reads occur at the closest preceding synchronization call.

Rule 3: signal writes occur at the closest succeeding synchronization call.

Rule 4: message-passing operations may be shifted in time, issued in the same cycle, or overlapped - subject to a small set of constraints we will look at next.

That is the whole user-level model. Pipelined loops, direct inputs and memories extend it in controlled ways. The thing to remember: whatever HLS adds - extra FSM states, pipeline stages, late physical sampling - must still be explainable by this model at the process interfaces.

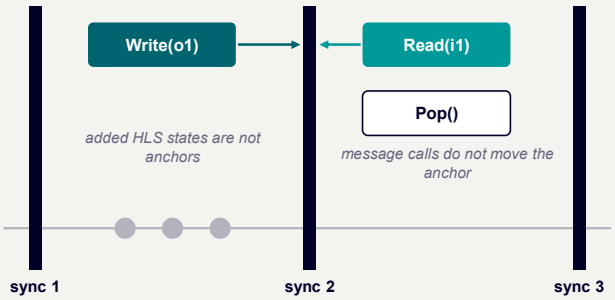
The four-rule summary here is for sequential processes. Combinational processes instead observe stable values after the network settles (R2C).

Signal IO is anchored to synchronization points

```
while (1) {
  o1.write(new_val);       // Write -> next wait
  do {
    wait();
  } while (!go.read());
  new_val = i1.read();     // Read -> prior wait
  new_val = f(new_val);    // no internal IO
}
```

- Read(sig) is anchored to the closest preceding sync call; Write(sig) to the closest succeeding one.
- Several writes before one anchor: the last one supplies the value.
- Same convention as Verilog: inputs sampled at the edge, outputs visible at the next edge.

One process, logical timeline



The physical sample may move; the logical Read or Write does not.

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For cycle-level reasoning, every signal access gets an anchor. A Read is anchored to the closest preceding synchronization call; a Write to the closest succeeding one.

In the code, o1.write is anchored to the wait in the do-while loop that follows it; i1.read is anchored to that same wait, which precedes it. The convention is exactly what Verilog engineers already know: inputs are sampled when the process wakes on the clock edge; outputs become visible at the next edge.

If several writes to the same signal share one succeeding anchor, the last one executed supplies the value; earlier ones are overwritten.

Two consequences matter for HLS. Extra FSM states that HLS inserts are never synchronization calls, so they do not create new anchors. And message operations between a signal access and its anchor do not move the anchor - which, as we will see under RULE 1, is exactly why you should keep signal IO adjacent to its wait.

Message passing: issue order and synchronization boundaries

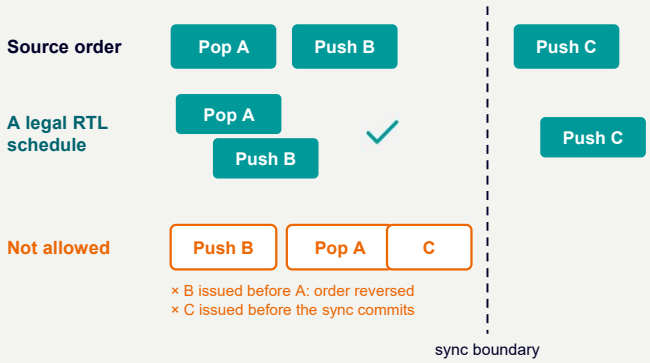
Before a sync
Issued no later than the cycle the sync commits; blocking calls complete by then.

After a sync
Issued only after the cycle in which the sync commits.

Same interface
Issued in source order — also across overlapped pipeline iterations.

Different interfaces
May overlap or share a cycle, but never issued in reverse source order.

Source order versus RTL issue



The sync is a fence: no earlier blocking request survives into the next interval, though a message and its sync may complete on the same edge.

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Rule 4 has four constraints.

Before a sync: every message operation in the interval must issue no later than the cycle in which that synchronization call commits - and blocking ones must have committed by then. It is a process-facing completion fence; the message and the sync may complete on the same edge.

After a sync: operations in the following interval issue only after the cycle in which the sync commits.

Same interface: always issued in source order, including across overlapped iterations of a pipelined loop.

Different interfaces: may be issued in the same cycle, and a later call may issue before an earlier one commits, but never in reverse sequence.

In the diagram, overlapping Pop A and Push B is legal. Issuing Push B before Pop A, or letting Push C slip in before the sync commits, is not. One subtlety: a buffered message that already completed at its process-facing boundary may still travel onward externally later.

What the no-reverse rule buys — and what it does not

Guaranteed by the default contract

HLS cannot introduce a new deadlock merely by reversing source-ordered message calls.

Same-interface order and distinct-interface no-reverse order both hold in the RTL, including across overlapped pipeline iterations.

Reversal — the factory deadlock — is ruled out by construction.

Not covered by this rule alone

Loop overlap, eager issue, finite capacities and the environment can still affect deadlock.

The separate completion result (Appendix K) handles these, under its capacity, correspondence and progress premises.

Its conclusion is acceptance of the required workload — not freedom from every internal wait cycle.

`STRICT_IO_SCHEDULING=relaxed`

Lets HLS reorder calls on distinct interfaces (never across sync). May add deadlocks; the formal results assume it is off.

1. **Run relaxed to see the order HLS prefers for QOR**
2. **Rewrite the source in that order**
3. **Return to the default scheduling mode**

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The most important default deadlock-avoidance rule is simple: HLS may overlap or shift message operations in time, but it may not reverse the dynamic source order of message-passing calls. Therefore HLS cannot introduce a new deadlock merely by reversing source calls.

Be precise about the limits. This is not a complete deadlock theorem. Overlapped loop iterations, eager issue, finite channel capacities and environment behavior can still affect deadlock. Those effects are handled by the separate completion result, whose conclusion is that the required workload is accepted - not that every internal wait cycle is gone.

There is a relaxed option, STRICT_IO_SCHEDULING=relaxed, that lets HLS reorder calls freely on distinct interfaces (never across sync calls). It can introduce deadlocks, and the formal results assume it is not used. The recommended loop: run relaxed to see the order HLS prefers for QOR, rewrite the source in that order, and go back to the default.

What HLS may do — and why the interfaces stay compatible

Transformation	What HLS may do	Why interface behavior is preserved
Added states	Insert cycles anywhere in a process, e.g., to share resources	Qualified extra states are silent; values and required observable order are preserved
Loop pipelining	Overlap loop iterations to raise throughput	Stages are internal; channel order, issue order and sync boundaries still hold
Direct inputs	Sample stable signals late, without capture registers	The value is stable, so a late sample equals the value at the anchor
Memory reordering	Move array accesses when proven conflict-free	Logical memory behavior and sync fences are preserved at the declared boundary
Finite buffering	Implement channels with bounded storage	The reference model is configured with the same back-annotated capacities
Cycle-exact modes	`IMPLICIT_FSM=true` or `IO_MODE=FIXED`	Interface cycles identical to pre-HLS; the internal FSM is identical only with IMPLICIT_FSM

Default: HLS may add latency. Cycle-exact modes trade that freedom for identical interface timing.

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Each transformation must preserve the qualified interface contract. Added states may be silent steps, but the contract preserves required observable order, not the total listing order of an arbitrary source simulation. Independent events can commute. Timing-sensitive choices retain their own qualification.

Loop overlap preserves source Issue order, channel order and synchronization boundaries under the declared process/boundary evidence. Direct-input sampling preserves the anchored value. Memory transformations stay below a faithful declared boundary. The configured source reference includes actual storage and coupling.

IMPLICIT_FSM=true requires identical interface cycles and internal FSM behavior under its contract. IO_MODE=FIXED requires identical interface cycles while allowing a different internal implementation.

References: rules Additional States, Additional Options, Appendix G; guide §§2–3.

Pipelined loops

Overlapped iterations, initiation interval 1



Orange lines: synchronization calls stay outside the pipelined loop body — before and after it.

No sync inside the loop body

No wait, ac_wait, SyncChannel or ac_sync in a pipelined loop. Put them outside, or don't pipeline that loop.

Message issues stay ordered

Last issue in one iteration ≤ first in the next. In a regular schedule, the message-issue span is at most one II.

Same-interface order stays intact

Across iterations, preserve access order on each message, signal or sync interface. Signal IO keeps its anchors.

Changes are flagged

If pipelining would reorder signal IO or split same-edge IO, the tool requires an explicit pragma (Cat7).

Pipelining: preserve the qualified interface contract. Finite storage and controller progress still need their own evidence.

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The timeline shows iterations overlapping in physical pipeline stages. It does not assign source-level message calls to those stages, and it does not by itself establish a sustained initiation interval of one cycle.

An ordinary pipelined loop body contains no explicit wait, ac_wait, SyncChannel or ac_sync. Place synchronization before and after the loop. Signal accesses remain tied to those real synchronization points, even when the physical accesses occur in different pipeline stages.

Source message requests remain ordered across iterations. The last message issue of iteration j cannot follow the first message issue of iteration j+1. In a regular repeating issue schedule, the span from an iteration's first to last message issue is therefore at most one initiation interval. The boundary may share a cycle. This limits message issues, not the entire computation or all completions (E3).

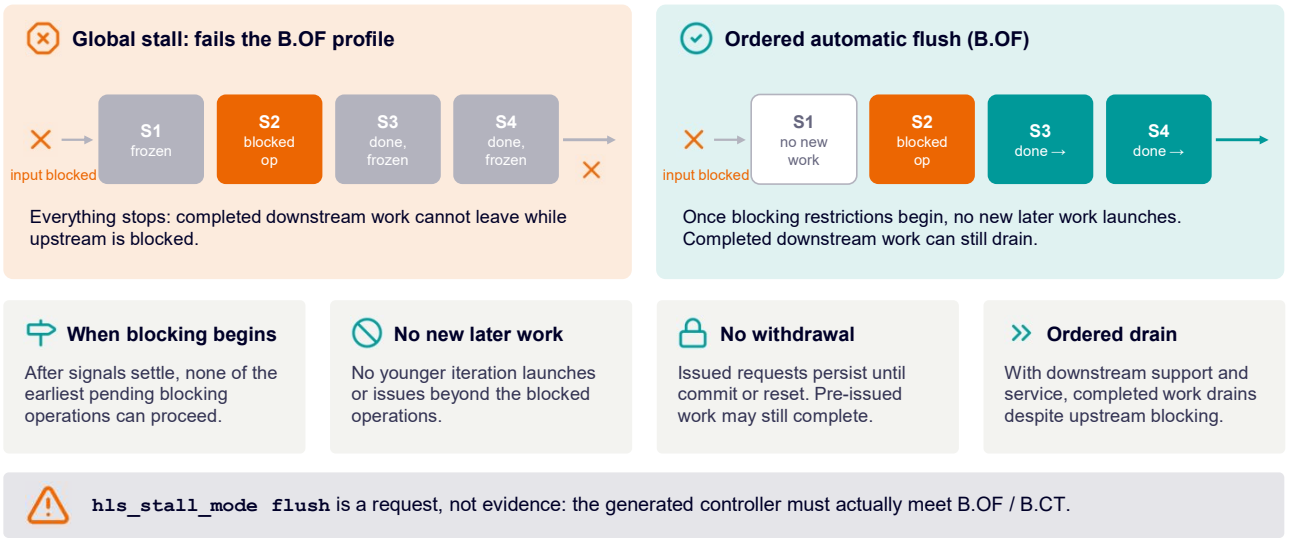
The same-interface rule is broader: accesses to each message, signal or synchronization interface keep their order across iterations. This does not authorize synchronization inside the pipelined body.

A physical FIFO read or forwarding step is not automatically the source Pop's issue or completion. The interface mapping must identify those events, and storage holding accepted messages still counts as communication storage (B.BoundaryBinding).

If pipelining would reorder signal IO or split IO that shared a source edge, the tool requires explicit authorization (Cat7). That authorization does not waive the observation rules or the separate controller, capacity and progress evidence.

(References: Pipelined Loops, Appendix B, E3, Core.RegSeq and guide §9.)

Pipelines must flush, not freeze



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The blocking restrictions begin at the first settled snapshot where there are pending blocking operations at the earliest unfinished source positions and none can proceed. These operations may form a group rather than a single call (the pending blocking frontier, Core.16).

While that condition persists, no new source-later evaluation, preparation or request may start beyond those operations. Requests legally started during the earlier preparation window are not retroactively forbidden.

Two kinds of activity may continue. Work already admitted downstream may drain. A later request issued legally before blocking began may also complete, provided it preserves the operands, descriptors, storage and effects still needed by the stalled older work. An affecting reset follows its declared treatment of outstanding work.

For completion, the controller must do more than obey the restrictions. Completed downstream work that is able to advance must receive ordered opportunities to leave, independently of upstream blocking or new input, under the declared downstream and service conditions (B.OF). This does not eliminate downstream backpressure or require every implementation to compact bubbles in the same way.

A global stall that indefinitely freezes required completed downstream work fails this completion profile. A flush pragma requests the behavior; evidence about the generated controller must establish it. An implementation outside the standard profile needs a separate proof that required requests become possible and progress.

(References: Core.16, Appendix B B.OF/B.CT and guide §§3 and 9.)

Direct inputs: stable signals, sampled late

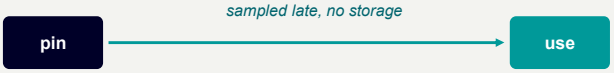
Ordinary signal read



Preserve the anchored value through added states or stages. This can require capture and pipeline storage.

```
#pragma hls_direct_input
```

sampled late, no storage



Logically, the Read still happens at its anchor. Physically, HLS may sample the stable signal whenever it needs the value.

The contract

The environment holds the signal stable once the receiving processes leave reset, until the next reset affecting them.

The freedom

HLS samples physically as late as it likes, with no capture registers — saving area.

The observation

Late samples are internal: no extra Read events, and the attributed value cannot change.

Dynamic resets

Allowed. Each reset starts a new stability epoch for the affected logic and its direct inputs.

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Ordinary signal reads happen at the closest preceding synchronization call, in both models. So if HLS adds states or pipelines the process, it must preserve the anchored value until it is consumed, which can require capture and pipeline storage. With many signals that is expensive, and often unnecessary.

The simplest fix is for signals that are held stable after the process comes out of reset - configuration registers, for example. Mark them with pragma hls_direct_input. HLS may then sample the physical signal as late as it likes with no register storage, while for scheduling and source/RTL comparison the Read is still treated as happening at its anchor.

The obligation moves to the environment: it must hold the signal stable within each reset epoch. Dynamic resets are fine; each reset starts a new stability epoch.

Later physical samples are internal implementation activity: they create no extra Read events and cannot change the value attributed to the logical Read.

Changing configuration safely: hls_direct_input_sync

```
#pragma hls_direct_input
sc_in<uint32_t> direct_inputs[N];
...
while (1) {
  #pragma hls_direct_input_sync all
  sync_in.sync_in();     // update point
  #pragma hls_pipeline_init_interval 1
  #pragma hls_stall_mode flush
  for (x = 0; x < direct_inputs[0]; x++)
    for (y = 0; y < direct_inputs[1]; y++) {
      sum = 0;
      for (s = 0; s < num_samples; s++)
        sum += sample_in[s].Pop()
               * direct_inputs[2 + s];
      ...  out1.Push(result);
    }
}
```

Abridged from Catapult Matchlib example 61

Only sync may gate these updates — never Push/Pop or ac_channel.

1. **DUT asserts and holds sync rdy**

   Only once old values are no longer needed — pipeline ramped down, or remaining uses protected.

2. **TB changes inputs, asserts vld**

   After seeing rdy; the new values settle before the sampling edge.

3. **Sync completes at the edge**

   With rdy held high, one cycle of vld is enough if that edge completes the handshake. Hold offers until commit or reset.

4. **Inputs stay stable**

   Until the next permitted update or an affecting reset.

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Sometimes control inputs must change while the design runs - for example, an image pipeline whose processing parameters change at the start of each frame. Before the inputs change, every remaining use of the old values must be finished or protected. Ramping down the pipeline is one way to do that; it is not required when the remaining old operands are retained and isolated from the pin change. Separately, capacity for next-frame work is withheld while the DUT is pending at the sync (Core.16(7)), but that admission control does not itself protect old-value users.

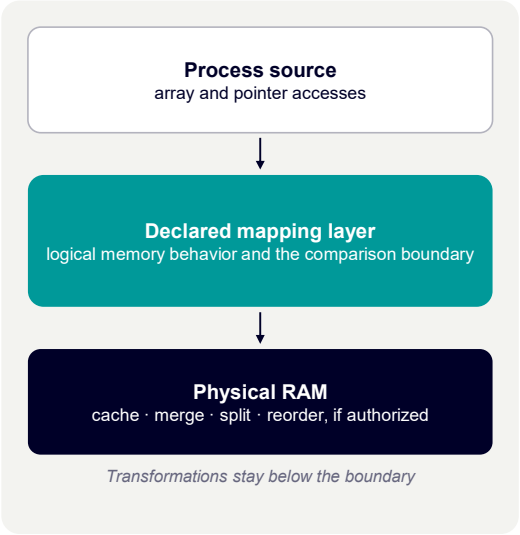
The mechanism is a SyncChannel or ac_sync call, tied to the direct inputs it controls by pragma hls_direct_input_sync. This is Catapult example 61.

The handshake: the testbench waits for the DUT's sync ready. By asserting ready, the DUT has finished with the old values or protected any remaining uses. The testbench changes the inputs and asserts sync valid, letting them settle before the sampling edge. With ready still high, the sync completes on that edge, so one cycle of valid is enough - but do not drop valid just because a cycle passed without a handshake. Then keep the new values stable until the next permitted update or reset.

Only the sync operation may gate these updates - not Push, Pop or ac_channel. If the environment needs to decide when to update, send a separate request asking the DUT to call sync; that request is not itself permission to change the inputs.

Both sync endpoints retain their offered signal and occurrence identity through commit or an affecting reset. Old-value protection begins before ready is asserted and remains valid while the offer is pending and for any later protected use. All eligibility and fence conditions act before the sampled overlap.

Arrays and memories



Declare the mapping

Name the logical memory behavior and the boundary at which source and RTL are compared.

Transform only below it

No visible interaction created, dropped, duplicated, relabeled or illegally reordered.

Respect sync fences

Accesses before a sync commit no later than the sync; never merge accesses across one.

Expose cross-process values

Use covered messages or anchored signals for shared values (J.Mem). Sync orders accesses but does not carry values.

Count backpressure

A memory that can stall a process belongs in the completion qualification.

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Arrays may remain arrays through synthesis and map to RAMs. Pointer accesses resolve to array accesses. Each mapped memory needs a declared description of its logical behavior and the boundary at which source and RTL are compared.

Below that boundary, authorized caching, merging, splitting and reordering may change physical RAM traffic. They must not create, drop, duplicate, relabel or illegally reorder interactions visible at the comparison boundary.

Synchronization still fences accesses. Earlier accesses commit no later than the sync, later accesses commit after it, and one merged commit cannot represent accesses on both sides.

When one process's write affects another process's values, decisions or operation order, the model must expose that dependency. The supplied proof route represents these values through covered messages and/or signals tied to synchronization points (J.Mem). Synchronization orders accesses but does not itself carry the written value or prove equal read values.

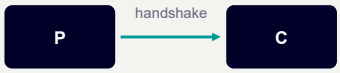
Reusable shared-memory wrappers, such as qualified ping-pong memories or native RAM FIFOs, can package this evidence. A more general proof based directly on shared memory state needs a separately supplied extension. A memory that stalls a process also needs to be included in the completion argument.

(References: Scheduling of Array Accesses, J.Mem and Core.MemFaithful.)

Channel capacity belongs in the reference model

Rendezvous

`B(c) = 0`



Push and Pop commit together: same cycle, same value.

Bounded FIFO

`B(c) > 0`



Pop needs an item at the start of the cycle; Push needs space then. No fall-through within a cycle.

Elaborated structure

joins, staging, gates



Explicit finite storage plus stateless ready/valid couplers, when one depth is not faithful.

The reference model is not the raw simulation

Raw SystemC
unadjusted channel depths

≠

Reference model Sys_ref
source + testbench with back-annotated capacities, arbitration and coupling

Matchlib note

In<> ports default to a one-place skid buffer for throughput modeling; check your version.

For final verification, match ports and capacities to the selected RTL and re-establish faithful capacities, e.g. with

`-DFORCE_AUTO_PORT=Connections::SYN_PORT`

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Channels come in two basic flavors.

Capacity zero is a rendezvous: the matching Push and Pop commit together in the same cycle with the same value.

Positive capacity is a bounded FIFO with cycle-boundary, non-fall-through semantics. A Pop needs an item present at the start of the cycle; a Push needs free space at the start of the cycle. A full FIFO cannot accept a Push just because a Pop happens on the same edge, and an empty FIFO cannot satisfy a Pop with that edge's new Push.

When a single depth does not describe the structure - joins, staging, coupled multi-input pipelines, sync gates - the reference model is elaborated with explicit storage plus stateless ready/valid couplers.

The important consequence: the reference model is not the raw SystemC simulation. It is the source and testbench configured with the implementation's back-annotated capacities and communication structure. Matchlib In ports model a one-place skid buffer by default for throughput modeling. Check that description against the library version and configuration in use; it does not by itself establish throughput accuracy. For final verification, configure ports and capacities to match the selected RTL.

RULE 1: every signal access needs one unambiguous anchor

Risky **A blocking Pop splits read from wait**

```
do { wait(); } while (!go.read());
int pop_val = pop1.Pop(); // may stall
new_val = i1.read();  // i1 may have
                      // changed by now
```

Fix: read i1 immediately after its wait. No cost in QOR or simulation speed.

Ill-formed **One write, two possible anchors**

```
wait();                  // WAIT 1
o1.write(some_value);
if (go.read()) { ... }
else { wait(); }         // WAIT 2
some_other_function();
wait();                  // WAIT 3
```

o1 binds to WAIT 2 or WAIT 3 depending on go. Flag as an error; move the write next to its intended wait.

RULE 1 Keep signal IO next to its wait. Whether an observable read occurs, and its value, must be determined at its anchor. Resolve or explicitly justify violations.

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For sequential processes, keep signal IO next to its synchronization call. In the left example, raw simulation may sample i1 after a stalled Pop even though the logical read belongs to the preceding wait. Moving the read next to that wait removes this mismatch risk. The right example has two possible following anchors for its write and needs a rewrite or an explicit justification allowed by the rules.

For a read included in the observable interface history, both whether it occurs and its value must be determined at its anchor. Consider wait(); x = in.Pop(); if (x) y = cfg.read();. A stable cfg value is not enough: the later Pop result decides whether the read exists. It is too late to add that read to a synchronization group that has already completed. Conditional reads are allowed when the decision is already determined at the anchor (retained-Read qualification).

There is a separately qualified way to observe configuration updates for designated direct inputs instead of observing ordinary reads (configuration-observation route). It has its own conditions and does not waive the ordinary read rules.

Resolve the diagnostic, supply the permitted explicit justification, or exclude the unsupported case from the guarantee.

(References: RULE 1, Core.17 and guide §5.)

Rules: RULE 2 (Cat4)

RULE 2: signal IO inside rolled loops

```
wait();             // start of while loop
#pragma hls_unroll no
for (int i = 0; i < 10; i++) {
  o1.write(i1.read() * i); // no anchor
}                          // in the loop
```

Why it is a problem

- With no sync inside the loop, every iteration shares one pair of anchors.
- Logically there is one Read; only the final Write supplies the value at the next anchor.
- An RTL that samples a fresh i1 per iteration differs from the anchored meaning. Added states never create anchors.

Per-iteration IO, loop not pipelined

Place a wait() in the loop body — and follow RULE 1.

One value intended

Move the read before the loop to make the intent explicit.

Loop will be pipelined

No in-body wait allowed: move IO outside, use message passing, use hls_direct_input(_sync), or don't pipeline.

RULE 2 Signal IO in a rolled loop needs a wait in the loop body.

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RULE 2 covers signal IO inside rolled, or preserved, loops.

In this example the loop body reads i1 and writes o1 ten times, but there is no synchronization call inside the loop. So every iteration shares the same anchors: logically, one read at the preceding wait, and only the final write supplies the value at the succeeding anchor. An RTL that samples a fresh i1 on each iteration across added clock cycles - or exposes each o1 value - differs from both the raw zero-time simulation and the anchored meaning. Added implementation states never create new anchors.

So decide what you meant. If you want per-iteration interaction and the loop is not pipelined, put a wait in the loop body and follow RULE 1. If you meant one value, read it before the loop. If the loop is to be pipelined, an in-body wait is not allowed: move the IO out, use message passing, use direct inputs, or do not pipeline that loop.

As with RULE 1, a warning must be resolved, explicitly justified, or treated as excluding the model from the formal guarantees.

Three common sources of pre-HLS / post-HLS mismatch

Non-blocking calls

PushNB / PopNB / nb_read / nb_write outcomes depend on latency — which HLS changes.

Instead

Use only when needed, e.g., time-based arbitration at a shared resource.

One-way messages as sync

Latency and storage between processes vary, so one channel does not synchronize two processes.

Instead

Use SyncChannel, or two opposite channels (AXI ar/r, aw/b) — weaker, but valid.

Signals without a handshake

wait()-synchronized signal IO without a two-way handshake is likely latency-sensitive.

Instead: a two-way handshake

```
out_dat = value;
out_vld = 1;
do {
  wait();
} while (out_rdy != 1);
out_vld = 0;
```

Rule of thumb: if behavior depends on how long something took, it is latency-sensitive — synchronize explicitly or isolate it.

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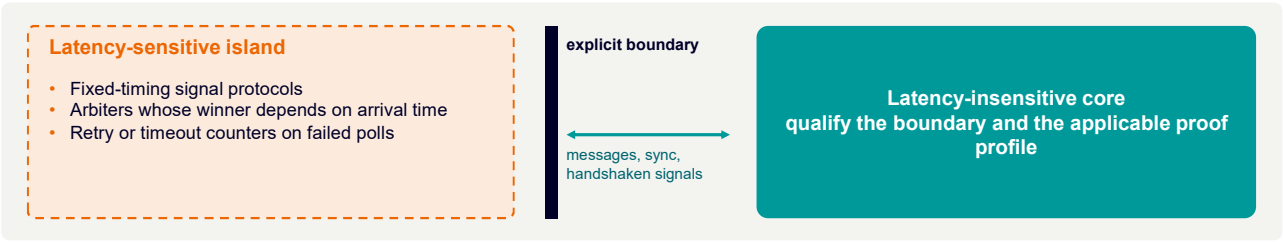
Three patterns cause most mismatches between pre-HLS and post-HLS simulation.

First, non-blocking message calls - PushNB, PopNB, nb_read, nb_write. Their outcome depends on latency, and latency is exactly what HLS changes. Use them only when nothing else works - time-based arbitration of several streams at a shared resource is the classic legitimate case.

Second, using a one-way message channel as if it synchronized two processes. Message latency and storage between processes vary, so it does not. Two channels in opposite directions can synchronize - think AXI ar and r, or aw and b - though that is weaker than an explicit SyncChannel or a two-way signal handshake.

Third, signal IO synchronized with a wait but without a proper two-way handshake. That is likely latency-sensitive. Use a handshake like the one shown; protocols such as APB use different but still latency-insensitive handshakes.

Isolating latency-sensitive behavior



Transactors

App. D

One side handles detailed signal timing (cycle-accurate SystemC); the other exchanges messages. See example 53_transactor_modeling.

One-way handshakes

App. E

Use only where back-pressure provably cannot occur, e.g., fixed-rate dat/vld streams (example 32). Assertions check only the runs exercised.

Snooping

App. L

Record relevant RTL observations, then replay them in order through a qualified source-side alignment harness.

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Timing-sensitive protocols, arrival-dependent arbitration and retry or timeout logic need explicit interface boundaries and supporting evidence. Isolating them is useful architecture, but does not automatically prove safety or completion.

Transactors use cycle-accurate signal behavior on one side and messages on the other. Example 53 illustrates this style. A one-way handshake needs an established no-backpressure property over the qualified operating, stall and reset conditions. Runtime assertions monitor only the runs actually exercised; a clean finite run does not prove that property.

Snooping can make a source run follow choices observed in a particular RTL run. The checking harness records each relevant RTL occurrence and supplies those observations to the source in order. It checks occurrence identity, operation kind, payload and any outstanding events. Simply AND-gating live signals can lose or confuse events when the source lags.

Keep the RTL free-running. Any throttling of the source must remain legal under its original protocol, and an issued source request must persist until commit or reset. The harness cannot suppress a mandatory handshake, invent a non-blocking failure or substitute a different payload (Appendix L harness requirements, L.Dir and W1–W5).

Snooping is one available technique, not a universal requirement. Matching one observed RTL run does not prove that all executions needed by the workload-completion claim make the same relevant choices and consume the same values. Nor does it establish progress.

(References: Appendix L, L.WC and guide §10.)

Rules: Appendix F

Modeling guidelines at a glance

In general

- Prefer `Connections::In/Out` + `SyncChannel` over signal IO and wait()
- Prefer `Pop()/Push()` over `PopNB()/PushNB()`
- For pipelined loops in completion claims, check that the generated controller meets the flush laws
- Unless modeling cycle-accurately, expect latency to differ between pre- and post-HLS models

When using signal IO

- Cycle-accurate process: use `disable_spawn` and the example-53 transactor style, with no Push/Pop
- Direct inputs: `hls_direct_input` (+ `_sync`); change only at allowed times
- Mixing signals with In/Out: use a real signal handshake, not the ports, for synchronization
- Keep signal IO right next to its wait()

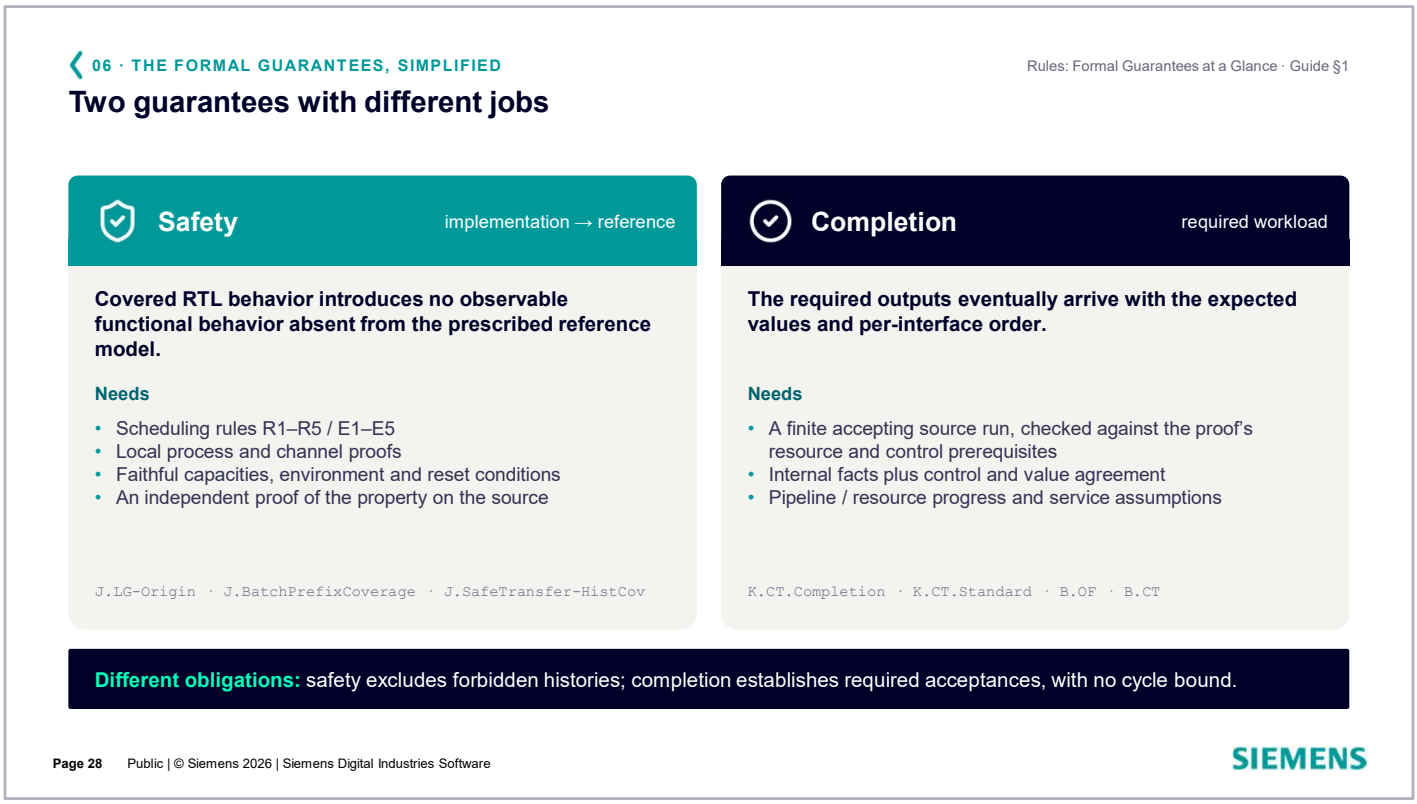
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Appendix F condenses the modeling practice into a short list.

In general: prefer Connections In and Out ports plus SyncChannel over raw signal IO and wait. Prefer blocking Pop and Push over the non-blocking variants. For pipelined loops used in completion claims, check that the generated controller really meets the flush laws - the directive alone is not evidence. And unless you are modeling a cycle-accurate process, expect latency to differ between the pre-HLS and post-HLS models; write your testbench accordingly.

When you do use signal IO: for a cycle-accurate process, use disable_spawn and the transactor style of example 53, with no Push or Pop in that process. For direct inputs, use hls_direct_input and possibly hls_direct_input_sync, and change the signal only at allowed times. When mixing signal IO with Connections ports in one process, use a proper signal handshake that does not rely on the ports for synchronization, and keep the signal IO right next to its wait.



The formal architecture separates safety from liveness, and it is worth keeping them apart in your head.

Safety runs from the implementation to the reference: covered RTL behavior introduces no observable functional behavior that is absent from the prescribed reference model. It is largely compositional - local evidence for processes, channels, atomic interactions, environment and reset is combined by a generic theorem. Its payoff is that properties established at the source hold on the RTL.

Completion is about getting work done: the required outputs eventually arrive with the expected values and per-interface order. It needs more: a finite accepting source run checked against the proof's resource and control prerequisites, correspondence of internal facts, pipeline and resource progress, and service or fairness assumptions.

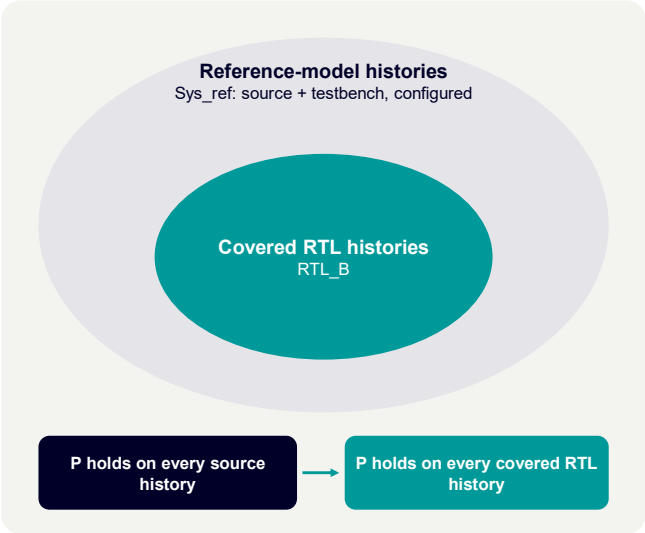
Neither implies the other. Safety does not promise any output at all; completion does not promise a cycle bound. Extra outputs are excluded by safety, not by completion.

The witness may be ordinary serial or a qualified group-serial witness for indivisible atomic groups. Completion agreement includes operation identities and consumed control decisions, not payloads alone.

A passing serial (Policy S) run is the starting point, not the whole check. The standard completion proof (Appendix K) also requires its complete prerequisite graph, including guaranteed-space dependencies, to be cycle-free; the completion workflow later in the talk explains this check (K.CT.SerialWitness, K.CT.Closure).

Rules: App. G · J.ReachableHistCov · J.SafeTransfer-HistCov

Safety: every covered RTL behavior has a source explanation



Direction matters

RTL → source. A particular pre-HLS run is not guaranteed to be reproduced by the RTL.

"Equal" means equal history

Same events, values, required order and indivisible groups. Independent events may occur in different cycles.

Write properties on that history

Check accepted values and required order, observing whole atomic groups. Exact cycle-count checks do not transfer.

The source property is separate

It must hold for all admitted source behaviors. One passing simulation is not that proof.

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Think of the interface histories that each model can produce. For every finite covered RTL execution, the safety proof constructs a legal reachable source execution with the same selected observable history. The direction is RTL to source. It does not promise that the RTL reproduces an arbitrarily chosen source simulation.

Equal history means equal events, values and required order at the selected interfaces. Independent events may occur in different cycles or be logged in a different order. Complete atomic groups remain together.

A scoreboard that checks accepted values and required request/response order can carry over if it uses only this history. A check requiring a response on an exact cycle generally cannot. Nor can a checker depend on which callback happens to run first for independent events, or inspect an intermediate logger state partway through an indivisible group (Batch observation profile).

The source property must separately hold for every admitted source behavior. One passing simulation does not establish that universal property. Once the source proof and the correspondence evidence are supplied, the property also holds on covered RTL executions.

Source issue order on different interfaces does not automatically create a strict order between their completions.

(References: Appendix G, J.ReachableHistCov and J.SafeTransfer-HistCov.)

Rules: Appendix G

The trace rules: R1–R5 for the source, E1–E5 for the RTL

Topic	Source rule (pre-HLS reference)	Implementation rule (post-HLS RTL)
Sync order	R1 · Sync calls occur in program order	E1 · Per-process sync order preserved
Signal anchoring	R2 · Reads at the preceding sync, writes at the succeeding sync. R2C: combinational logic observed at its settled fixed point	E2 · Signal visibility preserved at the same anchors
Isolation around sync	R3 · Messages stay in their interval; blocking ones complete by the sync	E5 · Messages cannot cross their immediate sync boundary
Message issue order	R4 · Never issued in reverse source order	E3 · Same ordering, prefix-closed, also across overlapped iterations
Channel semantics	R5 · Rendezvous (B = 0) or bounded FIFO (B > 0)	E4 · No drops, duplicates or reordering; occupancy stays within 0…B

τ_ref ≈ τ_post is an ordered compatibility relation — source witness on the left, RTL on the right — not a symmetric equivalence.

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Appendix G restates the contract as trace rules. The R rules describe the source reference; the E rules are the matching obligations the implementation must preserve.

Synchronization order: R1 and E1 - sync calls stay in program order within each process.

Signal anchoring: R2 and E2 - reads at the preceding sync, writes at the succeeding sync. Combinational processes are instead observed after their signals settle each cycle (R2C).

Isolation around sync: R3 and E5 - messages cannot cross their immediate synchronization boundary, and blocking ones complete by the sync.

Message issue order: R4 and E3 - never reverse source order. The rule applies throughout execution: if a later call has issued, every earlier one has too, including across overlapped iterations.

Channel semantics: R5 and E4 - rendezvous or bounded FIFO, no drops, duplicates or reordering, occupancy always between zero and the capacity.

Note the compatibility relation is ordered - source witness on the left, RTL on the right. It is not claimed to be a symmetric equivalence.

Rules: App. J · Guide §4

How local proofs compose into a system result

1 Local evidence

Per process, channel and atomic interaction: source and RTL steps correspond.

`contracts C0–C5`

2 Integration checks

The ready/valid network settles; environment, reset and capacities are faithful.

`settling · environment`

3 Generic composition

Induction over actual RTL steps builds a legal source prefix for every RTL prefix.

`J.LG-Origin · J.BatchPrefixCoverage`

4 Safety transfer

A property proved separately on the source carries over to the RTL.

`J.SafeTransfer-HistCov`

Why it matters

Local evidence replaces an intractable whole-system proof. For ordinary profiles, no design-specific global invariant is needed (J.OrdinaryConstructiveComposition).

What it does not assume

No fairness and no future cycles: deadlocked and terminated prefixes are covered too. It gives safety only — no progress conclusion.

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The safety proof builds a system argument from local evidence.

First, establish how each process, channel and atomic interaction implements its source behavior. The registered message profile states these local requirements explicitly (contracts C0–C5).

Next, check the integration: the ready/valid network settles, and the environment, reset behavior and capacities match the model.

The generic theorem then follows the actual RTL steps and constructs matching legal source steps. It covers every finite prefix and preserves complete observable groups (J.ConstructiveEdgeInduction, J.LG-Origin and J.BatchPrefixCoverage).

Finally, a property proved independently for all admitted source histories also holds for covered RTL histories (J.SafeTransfer-HistCov).

The benefit is reusable process-by-process and channel-by-channel evidence. For the ordinary profiles, the composition does not require an additional design-specific global invariant beyond its stated premises. It also does not assume future progress or fairness, so its safety conclusion includes executions that stop or deadlock.

(References: Appendix J and guide safety construction.)

Testbenches, properties and stimulus classes

The testbench is in the system

Reactive responses that depend on earlier DUT results are covered by the same joint argument.

One proof for a class of stimuli

A reusable correspondence proof can cover a declared class of inputs and choices; each new stimulus needs only binding and membership checks.

The source property is your job

It must hold for all admitted source behaviors. A passing sample campaign does not establish that.

Latency-sensitive testbenches

Branches, watchdogs or pass/fail decisions on raw cycle counts are not covered automatically: isolate, make explicit, or check at RTL.

Theorems: J.UniformConstructiveCoverage, J.ConstructiveClassSafe. Reuse can cut effort across a large, stable campaign — with no unconditional cost advantage.

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This slide is for the DV engineers in the room.

The testbench is normally included with the design in the compared system. That means reactive responses - stimulus that depends on earlier DUT results - are covered by the same joint argument. The implementation-side testbench need not be synthesized, but its runtime and interface semantics must be qualified.

One reusable correspondence proof can cover a declared class of stimuli and choices, not just a single test. Once it is established uniformly, each admitted stimulus needs only exact instance binding and membership checks.

But the source property is still your job. It must hold for all admitted source behaviors. A passing sample campaign or a shared testbench program does not establish that.

And latency-sensitive testbenches are a special case: branches, watchdogs or pass/fail decisions that depend on raw cycle counts are not automatically covered. Isolate them, make them explicit, or check them at RTL.

Reuse can reduce total effort across a large, stable campaign, but the rules are careful to claim no unconditional cost advantage.

Rules: App. K (K.CT) · App. N · Guide §8

Completion: proving the required outputs arrive

1 Declare the workload

The testbench acceptances you require, with values and per-endpoint order.

2 Run a qualified witness

A finite accepting run with real capacities. Use group-serial issue for qualified atomic groups.

3 Check the witness

Include all prerequisites required by the selected proof, including guaranteed usable space. Check the full graph for cycles.

4 Bind to RTL

Match internal facts, control and values across the witness and RTL. Timing-dependent choices need alignment.

5 Show progress

Qualify controller and resource progress. Keep clocks and required environment service running.

`K.CT.Completion` Every covered RTL execution that satisfies the service assumptions eventually accepts the workload with the same values and per-endpoint order.

Serial = conservative

Within a sync interval, each blocking call issues in a later cycle than all earlier blocking commits.

Failure is inconclusive

A failed witness or cyclic graph is not a deadlock verdict.

Not claimed

No cycle bound; unrelated or post-completion deadlocks are outside the result.

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Declare which testbench receiving events are required, with their values and order on each endpoint. A DUT merely asserting an output is not acceptance. The witness is a finite legal run of the source and testbench configured for this implementation. It accepts the required workload but need not drain unrelated storage or finish trailing work.

In an ordinary serial witness, a later blocking call issues only after every earlier blocking call in the same process and synchronization interval committed in an earlier cycle. Independent processes may still run concurrently (Core.PolicyS, K.CT.SerialWitness). Operations that must happen together remain indivisible. Schedule those whole groups together or use another qualified witness; splitting them changes the model (K.CT.GroupSerialWitness).

Checking the witness means including every prerequisite required by the selected completion proof: earlier operations, endpoint order, guaranteed usable FIFO space, and the values and decisions on which the work depends. For the standard pipeline proof, include the required work in preceding stages and older iterations. Explain why iterations start and why operations are required or skipped, including branches not taken. These decisions are evidence, not additional transfer events (K.CT.Closure and, for the standard route, K.CT.CtrlSupport).

A passing serial (Policy S) run already supplies an ordering for the dependencies enforced by that simulation. The standard completion proof (Appendix K) requires checking that the complete prerequisite graph used by its controller and resource arguments is acyclic. If every required dependency follows the recorded execution order, that order establishes acyclicity. The check must still establish that no required prerequisite has been omitted.

Exact modeling of the RTL's storage does not remove the guaranteed-space dependencies; the next slide shows why.

These are checks on the current sufficient completion certificate. Failure to construct the required acyclic graph is inconclusive; it does not establish that the RTL actually deadlocks (K.CT.SerialWitness, K.CT.Closure).

Prove that the witness and every covered RTL execution agree on the relevant operation identities, decisions and values. Matching only final outputs, running a deterministic simulator or snooping one execution does not establish this agreement (J internal-fact correspondence and K.CT.Agreement).

The controller and resources must actually reach the required operations and hold their requests. Once settled physical ready and valid are both high at the applicable edge, transfer is mandatory. Fairness concerns obtaining opportunities, such as a resource grant, or separately stated internal service; it is not permission to ignore a handshake.

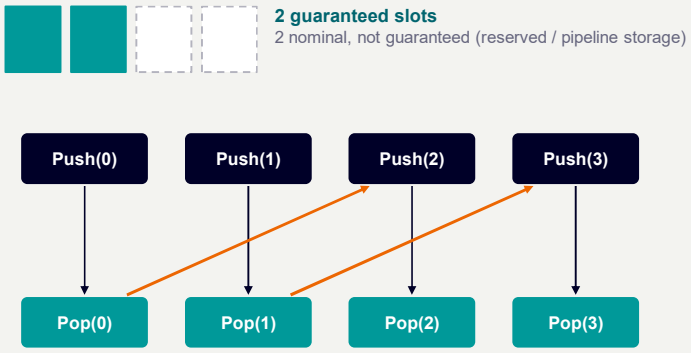
Keep required clocks and environment service running until acceptance. Reset, cancellation or test termination cannot remove an outstanding required obligation without proof that the obligation is preserved.

The result is eventual acceptance with matching values and endpoint order. It promises no cycle bound and no total deadlock freedom.

(References: K.CT.Instance, Witness, Closure, Agreement, Standard and Completion; Appendix N; guide §§7–9.)

Guaranteed capacity, not nominal capacity

Nominal capacity 4, guaranteed floor g = 2



Orange space edges: Pop(k − g) must precede Push(k) for k ≥ g — even if the source run used all four slots.

Use the floor

Witness space dependencies use the guaranteed floor g, never the nominal capacity.

Reserved space does not count

Reserved slots and storage inside a pipeline do not enlarge g automatically.

Watch k = g

The boundary case is the edge most easily lost.

Unbounded is not evidence

A run with unbounded channels says nothing about finite hardware.

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One detail of the completion workflow deserves an architect's attention: capacity.

The completion proof distinguishes nominal modeled storage from guaranteed usable space, called the floor g. Reserved slots, credits and storage inside a pipeline do not automatically enlarge that floor.

The prerequisite graph uses the floor. For an initially empty FIFO, Push k must precede Pop k, and Pop k minus g must precede Push k whenever k is at least g. Take nominal capacity four with a justified floor of two: Pop 0 must precede Push 2 and Pop 1 must precede Push 3 - even if the source run happened to use all four slots.

The boundary case k equals g is the edge most easily lost.

And an unbounded-channel source run is never evidence for finite hardware. If space edges introduce a dependency cycle, this sufficient check is inconclusive. More guaranteed space or a different proof route may be needed.

Why can a passing serial (Policy S) run still fail the standard completion proof's check? Consider a source reference that permits three data items to be accepted before any corresponding Pop, while the selected resource proof guarantees usable space for only two. A producer sends three items and then sends a start message on a separate channel. The receiver waits for that message before popping the data. A legal Policy S run can pass if the reference's actual storage and handshake laws permit those three initial sends.

With a guaranteed floor of two, however, the standard proof requires the first Pop before the third Push to establish space. That Pop requires the start message, and the start message follows the third Push. Those proof prerequisites form a cycle.

This does not show that the elaborated reference misrepresents the RTL, or that the RTL necessarily deadlocks. It shows that the selected two-slot resource guarantee is insufficient for this standard completion certificate. A stronger justified resource guarantee or another qualified proof route may resolve it. Exact modeling of the implementation does not automatically strengthen the resource guarantee being used.

If a separately justified run instead uses a two-slot capacity and the same source protocol, it stalls before accepting the required outputs, so the simulation reveals the obstruction directly. Using a smaller-capacity run as the witness is optional in general and requires the stated legal lift into the implementation-bound reference (K.CT.Capacity). The default exact elaborated model need not be replaced.

Illustrative example · Rules: E4, J and K.CT

Worked example: four values through a finite FIFO

Configured system

One producer sends 10, 20, 30, 40 through a two-slot FIFO to a testbench receiver.

The model

- Producer: four ordered blocking Push calls
- Testbench: four ordered blocking Pop calls
- One clock, initially empty FIFO, B = g = 2
- Fixed values; no timing-dependent branches

No other producers, hidden storage or affecting resets

The claims

The required workload is the testbench's acceptance of all four values in that order.

Two separate obligations

- Safety: accepted values form a prefix of the list
- Completion: all four acceptances eventually occur
- The source safety property is proved separately
- Progress needs controller and service evidence

The source and RTL may use different cycle schedules

A legal finite source run that accepts all four values at capacity B = 2 supplies the witness.

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Use one producer and one testbench receiver connected by an initially empty ordinary FIFO of capacity two. The producer makes four blocking Push calls with values 10, 20, 30, 40; the receiver makes four ordered blocking Pop calls. Both obey the registered-return and endpoint-order rules. There are no other producers, cancellations, affecting resets or timing-dependent branches. Both slots are usable, so B = g = 2.

The required workload is the four receiver acceptances, with those values in that order. The source safety property is proved separately: every accepted sequence is a prefix of the list, with no extra or duplicate values. The safety correspondence then carries that property to qualified RTL. A stalled run can satisfy the property without accepting all four values, which is why completion needs a separate argument.

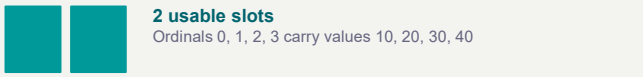
A legal finite accepting source run supplies the completion witness. Here it can alternate Push(0), Pop(0), Push(1), Pop(1), Push(2), Pop(2), Push(3), Pop(3), using successively later legal cycles. This is deliberately conservative; independent processes need not generally be serialized together. The run uses neither fall-through nor replacement of an item on the same edge that a full FIFO removes one.

One such run supplies the witness only. It does not by itself prove the universal source safety property, source/RTL agreement or controller progress. The next slide shows the prerequisite graph and the additional progress conditions.

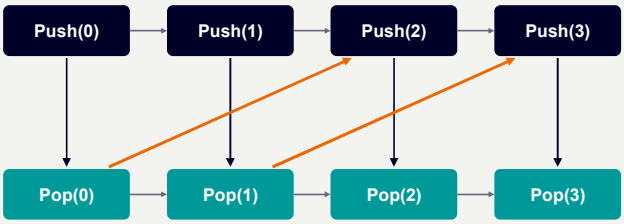
(References: Core.PolicyS, E4, J.SafeTransfer-HistCov and K.CT.Witness/Closure/Completion.)

Worked example: space, order and eventual acceptance

Initially empty FIFO: capacity B = floor g = 2



2 usable slots
Ordinals 0, 1, 2, 3 carry values 10, 20, 30, 40



Space prerequisites: Pop(0) before Push(2), Pop(1) before Push(3). Per-endpoint order also applies.

A legal finite witness

Push(0), Pop(0), Push(1), Pop(1), … on successive legal cycles accepts all four values.

No dependency cycle

Each item needs its Push. Reusing a slot needs an earlier Pop. Horizontal arrows preserve endpoint order.

What makes it progress

Finite preparation, persistent requests, continued clocks and receiver service expose each next transfer.

The conclusion

With source/RTL correspondence and progress proved, all four values are accepted in order. No cycle bound follows.

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The graph includes three kinds of prerequisite. Each Push must precede its matching Pop. Each endpoint keeps its own order, shown by the horizontal arrows. With two usable slots, Pop(0) must precede Push(2), and Pop(1) must precede Push(3), shown in orange.

The alternating witness respects every arrow. Its eight transfers supply the required supporting work, and its four Pops are the receiving events we require (the formal sets W and R). Values and decisions are fixed and do not depend on timing. This example therefore has no cycle in its prerequisite graph.

Actual RTL may use a different schedule. Once the predecessors are complete, the controller must finish the necessary local preparation and hold the next request. For this ordinary FIFO, the item or free slot must exist at the start of the cycle. A full FIFO cannot use a same-edge Pop to justify a Push, and an empty FIFO cannot use a same-edge Push to satisfy a Pop.

Continued applicable clock edges then force transfers whenever settled ready and valid are both high. There is no arbiter in this example and no separate selection decision after a handshake becomes enabled. Required preparation and receiver service still need evidence that they occur.

With the source/RTL correspondence, agreement and progress conditions established, follow the prerequisites through the finite graph: every required Pop eventually accepts its expected value. This gives no fixed cycle count or throughput guarantee.

(References: B.CT.FIFO, Core.TransferLaw, K.CT.Closure, Exposure, Service and Completion; guide §8.)

Completion scope and additional qualification

Standard completion

A safety-qualified feature still needs its applicable controller and progress evidence.

Within a declared profile

- Source-aligned control and agreed values
- Faithful finite resources and qualified staging
- Persistent requests and useful drainage
- A legal witness and complete prerequisites

`K.CT.Standard · B.OF · B.CT · Q.CT`

Additional qualification

Some structures need additional proof that their behavior and progress fit the completion claim.

Examples

- General arbiters or shared credit pools
- Timing-dependent polling or signal control
- Staging outside the named delivery profiles
- Reset crossing outstanding required work

`K.CT.Agreement · Q.CT.Extension`

Failure to meet one sufficient set of conditions is inconclusive; another proof route may apply.

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The standard completion result covers a stated set of implementation behaviors. The left column summarizes the needed evidence. It is not a blanket guarantee for every design that can be synthesized. The right column lists cases that need additional proof, not prohibited architectures.

For timing-dependent decisions, establish that the workload example and every relevant RTL-matched source execution make the same required choices and use the same values. This may follow from common timing-independent inputs and computations, or from a separate proof of that agreement (K.CT.Agreement).

General arbitration and shared credit pools need their own resource, request-progress and service rules. Supported input-staging arrangements also have specific requirements: the usual proof establishes placement and a persistent request without assuming that the target item has already arrived. If placement requires travelling with the item, the separate pre-request extension must justify that path (Q.CT.Extension).

A reset that crosses outstanding required work needs proof that the obligation is preserved. Atomic groups must remain indivisible and may need a witness that schedules whole groups together. Synchronization involving more than two participants and other unsupported combinations retain their additional obligations.

Failure to satisfy one sufficient profile is inconclusive. A different qualified proof route may still apply.

(References: K.CT.Standard, Explicit exclusions and extensions, K.CT.Agreement, Q.CT.Extension and guide §§7–9.)

What the guarantees do not claim

Reverse realization

The RTL need not reproduce an arbitrarily chosen source simulation.

Cycle equality

No identical timing, registers, latency or throughput — those need their own timing model and evidence.

A cycle bound

Completion means “eventually”, not “within N cycles”.

Total deadlock freedom

Deadlocks in unrelated work, or after the required outputs are accepted, are outside the result.

Automatic tool certification

The proofs do not certify a particular HLS run; the flow must supply the evidence.

Multiple clocks, async reset

One global clock per theorem instance; asynchronous reset assertion needs a separate argument.

When a sufficient check fails, the result is inconclusive — it does not prove the design wrong.

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It is just as important to know what is not claimed.

No reverse realization: the RTL need not reproduce an arbitrarily chosen source simulation. Appendix L can align a source run to a selected RTL run, but that is a verification technique, not a theorem that every source behavior is realizable.

No cycle equality: identical timing, registers, latency or throughput are not implied; they need their own timing model and evidence.

No cycle bound: completion is 'eventually'.

No total deadlock freedom: deadlocks in unrelated work or after the required outputs are accepted are outside the result.

No automatic tool certification: the proofs do not certify a particular HLS invocation or RTL instance; the flow must supply evidence. The same framework can apply to hand-written RTL if equivalent evidence is supplied.

Scope limits: one global clock per theorem instance - Appendix O only sketches multiclock - and asynchronous reset assertion needs a separate argument.

And when a sufficient check fails, the result is inconclusive; it does not prove the design wrong.

Dynamic reset may qualify for safety under its own profile. For completion, an affecting reset, cancellation or premature testbench termination cannot remove outstanding required work without a separate preservation qualification. Required clocks and environment service continue until acceptance.

Design constraints for architects

1 Anchor sequential signal IO
Preserve whether each observable read occurs, its value and its sync point.

2 Stable direct inputs
Stable per reset epoch, or updated only via the sync handshake.

3 Explicit timing sensitivity
Latency-dependent environment behavior isolated or made observable.

4 Polling is a design decision
Non-blocking control picks a proof route; prefer blocking calls.

5 No unbounded busy polling
Qualify retries under the chosen profile, or use a blocking call.

6 Expose shared-memory values
Each cross-process memory dependency via a supported boundary.

7 Model the real channel
No hidden fall-through, bypass, debug drain or side entrance.

8 Elaborate joined structures
Joins, coupled multi-input pipelines and sync gates modeled explicitly.

9 Reset with a disposition
Reset both channel ends together, or define a flush / abort protocol.

For completion: qualify arbiters and joint transfers, use real capacities, and maintain service. Reset or cancellation cannot erase required work without a preservation proof.

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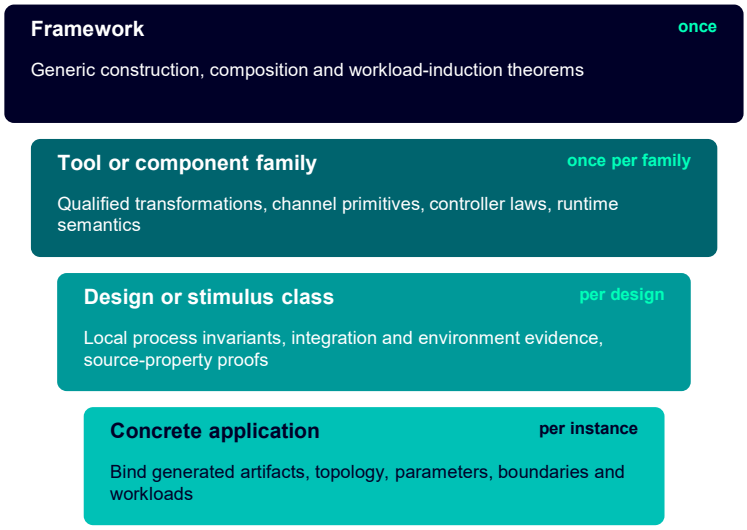
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Appendix F gathers the design choices that most directly decide whether the guarantees can be applied. A design can synthesize and work correctly and still sit outside the theorem scope if one of these is left implicit.

Briefly: give every signal access an unambiguous anchor. Keep direct inputs stable or update them only through the sync handshake. Make timing-sensitive environment behavior explicit. Treat non-blocking polling as an architectural decision about which proof route you are taking, and do not rely on unbounded busy polling. Expose cross-process shared-memory dependencies. Model the channel behavior the RTL actually provides - no hidden fall-through, bypass, debug drains or side entrances. Use an elaborated model for joins, coupled multi-input pipelines and sync gates. And give every dynamic reset a channel-consistent disposition - reset both ends together, or define an explicit flush or abort protocol.

For completion, add: point-to-point channels with explicit arbiter processes, explicit multi-lane transfers, persistent blocking requests, real capacities, and explicit barrier, end-of-stream and abort protocols.

Who supplies which evidence



The division of labor

- Architects choose interfaces, reset and communication structures that admit the correspondence.
- The tool and verification flow must supply or check the detailed evidence for the claimed scope.
- Evidence: documentation, lint and static checks, targeted RTL / interface checks, formal verification.
- Written with the precision intended to support mechanization in Lean.

Most of these conditions were always there. The formalization names them, so they can be audited instead of quietly assumed.

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The formal appendices are large, and it is fair to ask who has to deal with all of that.

The answer is layered. The framework - the generic construction, composition and workload-induction theorems - is proved once. A tool or component family, such as Catapult and the Matchlib primitives, is qualified once per family: transformations, channel primitives, controller laws, runtime semantics. A design or stimulus class supplies its local process invariants, integration and environment evidence, and the independent source-property proofs. A concrete application binds the generated artifacts, topology, parameters and workloads.

The architect's job is to choose interfaces, reset behavior and communication structures that admit the correspondence. The tool and verification flow construct and check the detailed records. Evidence can come from documentation, static checks and lint, targeted RTL or interface checks, or formal verification - and the appendices are precise enough to support mechanization in Lean.

Most of the engineering conditions were always there. The formalization just names them, so they can be audited instead of quietly assumed.

The evidence division is a responsibility model, not a claim that every record is currently generated automatically. Reusable qualifications bind the relevant tool/runtime versions, libraries, feature profiles and parameters. Changes outside that domain require renewed evidence.



The scheduling contract lets most functional verification and debugging happen at the source while HLS retains freedom to optimize. Its guarantees depend on the declared models, interfaces, implementation behavior and supporting evidence.

A source interface property proved for all admitted source behaviors also holds on covered RTL behaviors once the safety correspondence is established. Required-workload completion additionally needs a legal finite accepting source run, agreement on internal values and decisions, a complete prerequisite graph, controller and resource progress, and continuing service.

Architects choose communication, memory, reset and timing structures that support the intended claim. The tool and verification flow supply or check the detailed evidence for the supported configurations. The framework specifies those obligations; it does not automatically certify a particular tool flow.

The contract is tool-neutral and provides a starting point for standardization in the Accellera Synthesis Working Group.

(References: Formal Guarantees at a Glance, Appendices J, K and M; guide §§1 and 11.)

REFERENCES AND FURTHER READING

References and further reading

Repository github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG

Catapult HLS User View Scheduling Rules (25 October 2026)
.../matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

Formal Architecture Guide (25 October 2026)
.../matchlib_examples/doc/formal_architecture_guide.pdf

Brief summary of the scheduling rules
.../matchlib_examples/doc/brief_scheduling_rules.txt

Examples cited: 09, 12, 15, 32, 53, 61
.../matchlib_examples/examples

Making shift-left real (white paper)
hls.academy/making-shift-left-real-why-systemc-matchlib-and-high-level-synthesis-...

Verifying hardware at the source (white paper)
hls.academy/topics/systemc-modeling/verifying-hardware-at-the-source-...

Reading paths in the rules

Architects and HLS users
Main body through App. F; App. B and C; App. K for completion

Safety reviewers
Normative Formal Core; App. G; App. J constructive theorems

Overview and evidence
App. M formal results; M.7a evidence checklist

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All of the material is public in the Matchlib Examples Kit for the Accellera Synthesis Working Group on GitHub: the full rules document, the formal architecture guide, a brief summary of the rules, and the examples referenced in this talk.

For reading paths: HLS users and architects should read the main body through Appendix F, Appendices B and C for pipeline and request behavior, and Appendix K for the completion workflow. Safety reviewers should start with the Normative Formal Core, Appendix G and the constructive theorems in Appendix J. Appendix M gives a readable overview of the formal results, and M.7a is the evidence checklist.